## 1. ConvertAudioEngine.ps1 *audio-processing script*

### Downmixes AAC 7.1, EAC3 7.1, TrueHD 7.1, PCM/FLAC 7.1 to DDP 5.1 (1024k)

- Re-encodes **5.1 audio** to **DDP 5.1 (768k)**
- Encodes **stereo audio** to **DDP 2.0 (256k)**
- Passthrough for **EAC3 Atmos**, **EAC3 5.1**, **EAC3 2.0**, and **TrueHD 5.1**
- Maintains perfect sync on long-duration files
- Normalizes malformed channel layouts
- Removes 2-channel commentary tracks
- Produces clean, predictable, device-safe output

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## Requirements

- **PowerShell** 5.1 or 7.x (fully tested on both)
- **FFmpeg 8.1 or newer** — Must be in your PATH or the same folder as the script. You can grab the latest builds from the official FFmpeg download page.

1. Place `ConvertAudioEngine.ps1`, `ffmpeg.exe`, and `ffprobe.exe` in the same folder.
2. Run the script:

```
powershell -ExecutionPolicy Bypass -File .\ConvertAudioEngine.ps1 ".\YourMovie.mkv"
```

## 🎧 Supported Audio Codecs

The engine supports and processes the following codec families:

| Codec Family | Formats Included |
|---|---|
| **AAC** | AAC LC, HE-AAC, AAC 5.1, AAC 7.1 |
| **EAC3 / Atmos** | EAC3, EAC3-JOC (Atmos) |
| **TrueHD** | TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos |
| **DTS** | DTS Core, DTS-HD MA, DTS-HD HRA |
| **PCM** | pcm*s16le, pcm*s24le, pcm*f32le, pcm*f32be |
| **FLAC** | All FLAC channel layouts |

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## 2. FFmpeg Global Options:

These are global flags to ensure FFmpeg handles large, high-resolution files without data corruption or sync drift.

- `-threads 8` * User-adjustable (4–16), 0 = automatic
  - **What:** Sets the number of CPU threads the encoder is allowed to use.
    - **Why:** Helps balance encoding speed and system stability. It prevents CPU saturation that can lead to frame drops or system instability during the muxing process.
- `-analyzeduration 200M & -probesize 200M`
  - **What:** Increases the initial data analysis buffer to 200 Megabytes.
    - **Why:** High-bitrate audio like TrueHD with Atmos metadata or AAC 7.1 requires a deep probe to fully parse the channel layout. Standard 5MB, 10MB, 20MB probes often misidentify these tracks, causing the encode to fail or use the wrong channel map.
- `-err_detect ignore_err`
  - **What:** Instructs FFmpeg to bypass minor bitstream errors.
    - **Why:** Prevents a single corrupted audio frame (common in digital rips) from aborting a multi-hour conversion process.
- `-drc_scale 0`
  - **What:** Disables Dynamic Range Compression (DRC).
    - **Why:** Ensures the output audio maintains the full dynamic range of the source. It prevents the encoder from "flattening" the sound or lowering the volume of loud peaks.

## 3. The "De-Sync" Safeguard

These flags are applied after the input to maintain perfect alignment on long-duration files.

- `-avoid_negative_ts make_zero`
  - **What:** Forces all stream timestamps to start at absolute zero.
    - **Why:** Many digital files contain negative initial "Presentation Time Stamps" (PTS); this clamps them to zero to prevent audio lag or lead.
- `-max_muxing_queue_size 14000` *(1024-100000) recommended range
  - **What:** Increases the muxing buffer headroom.
    - **Why:** Audio down-mixing and re-encoding (AAC 7.1, TrueHD 7.1, ...) can run slower than the muxer, causing FFmpeg's queue to overflow and trigger errors. 14000 provides enough headroom to prevent these overflows on **all movie lengths.**
- `-map 0:v? -c:v copy`
  - **What:** Maps all video tracks and sets them to stream-copy mode.
    - **Why:** Ensures zero quality loss for the video while moving at maximum speed.
- `-map_metadata 0 -map_chapters 0`
  - **What:** Copies global metadata and chapter markers from the source.
    - **Why:** Preserves the original file's organizational structure (Title, Year, Scene markers).

## 4. Stereo (2.0) Optimization: Dolby Pro Logic II Compatibility

For stereo tracks, the engine uses specific flags to ensure they playback correctly on both standard speakers and surround sound systems.

- **-dsur_mode 1**
  - **What:** Enables the Dolby Surround Mode flag in the EAC3 bitstream.
    - **Why:** Tells the playback receiver that the 2.0 track contains matrixed surround information (like Pro Logic II), allowing it to properly expand the soundstage.
- **-stereo_rematrixing 1**
  - **What:** Explicitly enables rematrixing during the stereo encode.
    - **Why:** Ensures that the left and right channels are balanced correctly during the conversion from higher-channel sources to stereo.

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## 5. Encoding & Downmix Execution

These parameters determine how the audio is shaped and processed during encoding.

- `-filter:a:$i $panFilter`
  - **What:** Applies the custom channel-mapping matrix (the Pan Filter).
    - **Why:** Necessary for downmixing 7.1 or 5.1 to smaller layouts. It ensures that "lost" channels (like back-surrounds) are projected into the remaining speakers rather than simply deleted.
- `-c:a:$i eac3`
  - **What:** Sets the codec for audio stream `$i` to Dolby Digital Plus (EAC3).
    - **Why:** Chosen for its high efficiency and broad compatibility with modern smart TVs and home theater receivers.
- `-b:a:$i $t.Bitrate`
  - **What:** Sets the target bitrate (e.g., 1024k for 5.1, 256k for 2.0).
    - **Why:** High bitrates ensure the "transparency" of the encode, making it difficult to distinguish from the lossless original.
- `-dialnorm -31`
  - **What:** Sets Dialogue Normalization to the reference "off" position.
    - **Why:** Prevents the playback device from applying its own internal volume leveling, preserving the original mix's intended loudness.

- `-cutoff 20000`
  - **What:** Sets the low-pass filter frequency to 20kHz.
    - **Why:** Preserves the high-frequency "clarity" of the audio that standard encoders sometimes discard to save space.
    - **Applies to:** Downmix, 5.1 re-encode. Not applied to the stereo (2.0) encode path.
- `-metadata:s:a:$i "title=..."`
  - **What:** Injects a custom string into the stream title metadata.
    - **Why:** Allows users to see exactly what the track is (e.g., "DD+ 5.1 Downmix") in their media player's audio selection menu.

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## 6. Audio Rules & Priority Logic

The engine uses a **Priority Mapping (0-100)** to evaluate tracks based on quality and codec type.

## 7. Audio Rules & Priority Logic

### EAC3 / Atmos Rules (Priority 100)

| Condition | Action | Output | Notes |
|---|---|---|---|
| **Atmos (JOC)** | Passthrough | Original EAC3 Atmos | Preserves object metadata. |
| **EAC3 7.1** | **Downmix** | DDP 5.1 @ 1024k | Pri 99: Maps 8 channels to 6 using Pan Filter. |
| **EAC3 5.1** | Passthrough | Original EAC3 5.1 | Pri 98 |
| **EAC3 2.0** | Passthrough | Original EAC3 2.0 | Pri 97 |

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### TrueHD Rules

| Channels | Action | Output | Priority |
|---|---|---|---|
| 5.1 | Passthrough | Original | 81 |
| 7.1 | Downmix | DDP 5.1 @ 1024k | 80 |

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## AAC Rules

| Channels | Action | Output | Priority |
|---|---|---|---|
| 7.1 | Downmix | DDP 5.1 @ 1024k | 92 |
| 5.1 | Encode | DDP 5.1 @ 768k | 91 |
| 2.0 | Encode | DDP 2.0 @ 256k | 90 |

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## DTS Rules

| Variant | Action | Output | Priority |
|---|---|---|---|
| DTS-HD MA/HRA (multichannel) | Encode | DDP 5.1 @ 1024k | 73 |
| DTS Core (multichannel) | Encode | DDP 5.1 @ 768k | 72 |
| DTS-HD 2.0 | Encode | DDP 2.0 @ 384k | 71 |
| DTS 2.0 | Encode | DDP 2.0 @ 256k | 70 |

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## PCM / FLAC Rules

| Channels | Action | Output | Priority |
|---|---|---|---|
| 7.1 | Downmix | DDP 5.1 @ 1024k | 62 |
| 5.1 | Encode | DDP 5.1 @ 768k | 61 |
| 2.0 | Encode | DDP 2.0 @ 384k | 60 |

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## 8. Spatial Audio: The Pan Filter Matrix

When 7.1 audio is reduced to 5.1, standard downmixing often loses spatial detail from the side channels.

### The Formula

`aformat=channel_layouts=7.1,pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707*SL|BR=BR+0.707*SR`

### Layout Pinning (aformat)

**What:** Forces the decoder to output a strict 8-channel 7.1 layout *before* the math is applied.
**Why:** Atmos metadata or variable codecs can cause "amorphous" decoding where channel positions shift. **Pinning** ensures the Pan filter's "Side Left" label actually points to the side surround data.

### The Math (0.707)

**What:** Applies a **-3dB** coefficient to the Side Surrounds (SL/SR).
**Why:** Prevents digital clipping and ensures the side-channel sounds don't overpower the rear soundstage.

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## 9. Commentary Filtering Logic

**Only 2-channel commentary tracks**are removed to maximize space without risking primary content.

- **Detection Methods:** Title matching, keyword detection, and case-insensitive search.
- **Keywords:** `commentary, director, producer, writer, cast, behind, bonus, alt, interview`.
- **Why 5.1 Commentary is NOT removed:**
  - It is extremely rare in retail media.
    - It is often hard to tell apart from legitimate "Alternate Mixes" or "Home Theater Optimized" tracks.
    - Removing it carries a high risk of "False Positives" (deleting real movie content).

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## 10. Malformed Layout Guards

The script corrects illegal 7-channel layouts:

- **AAC 7ch** → 6ch, **EAC3 7ch** → 6ch, **TrueHD 7ch** → 8ch, **PCM/FLAC 7ch** → 8ch
- **Dialnorm (-31):** Sets Dialogue Normalization to -31dB. This is the "off" setting for Dolby metadata, preventing the playback device from applying its own "loudness leveling" and ensuring you hear the audio exactly as encoded.

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## ⛔ Fallback Rules

If no specific rule matches, the following logic is applied:

| Channels | Action | Bitrate |
|---|---|---|
| **≤2ch** | Encode | 256k |
| **3–6ch** | Encode | 768k |
| **>6ch** | Downmix | 1024k |

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## 11. Summary Engine

At = Audio processing Summary =, the script displays a table to audit the engine's work.

| Column | Description |
|---|---|
| **Index** | The original stream number (1, 2, 3,...). |
| **Codec** | The source format (e.g., TrueHD, DTS). |
| **Channels** | The detected channel count (e.g., 7.1, 5.1). |
| **Action** | **Green**: Passthrough, **Yellow**: Downmix, **Blue**: Encode, **Red**: Removed. |
| **Output** | The final codec and bitrate configuration. |
| **Priority** | The rank assigned by the engine (0-110). |
| **Rule** | The specific logic match (e.g., "TrueHD 7.1 Rule"). |